

SANNIDHYA RAY

Course : **B.E.**, Electronics & Communication Engineering and **M.Sc.**, Physics, 2028

Email : F20231131@PILANI.BITS-PILANI.AC.IN

Mobile : 9205733240

Social : [Linkedin](#) [Github](#)

CGPA : 7.28



ACADEMIC DETAILS

COURSE	SPECIALIZATION	INSTITUTE/COLLEGE	BOARD/UNIVERSITY	SCORE	YEAR
CLASS XII	PCM/CS	Delhi Public School, Greater Noida	CBSE	94.8 %	2023
CLASS X	General	Delhi Public School, Greater Noida	CBSE	96 %	2021

Subjects / Electives	Fundamentals of Finance and Accounting, Derivates and Risk Management, Physics of Semiconductor Devices
Technical Proficiency	Python, Machine Learning, Deep Learning, Full Stack Development, Object-Oriented Programming, C Programming, Linux, Bash, MySQL, Verilog, Assembly Language, Cadence Virtuoso, MATLAB, Quantum GIS, Google Earth Engine

INTERNSHIPS AND PROFESSIONAL EXPERIENCE

Summer Intern, STMicroelectronics	Jun 2026 - Present
- Built an end-to-end ML pipeline for multi-corner PVT characterization of SRAM bitcells, validated in Python (scikit-learn, NumPy) - Developed Gaussian Process Regression surrogates, reducing SPICE simulation time for PVT margin analysis by 50%	
Remote Sensing and GIS Intern, Department of Space	Jul 2025 - Aug 2025
- Utilized Google Earth Engine and Quantum GIS to process satellite imagery and analyse ecological trends over multiple decades - Analysed vegetation loss across ~10,000 sq. km of the Sundarbans by processing multi-decade satellite imagery (2000-2023)	
Summer Intern, Siemens	May 2025 - Jul 2025
- Cut inspection time by ~60% on wave soldering machines by building a monitoring system tracking 12+ process parameters - Reduced reporting turnaround by ~80% by automating 3 production lines via C#, Python (PyQT5, PyODBC) and SQL integrations	

PROJECTS AND PUBLICATIONS

TicTacToeTwo (tictactotwo.vercel.app) - An advanced web-based multiplayer game	Feb 2026 - May 2026
- Implemented ELO-ranked real-time matchmaking for a full-stack multiplayer game with Firestore-secured auth deployed on Vercel - Integrated Firestore Realtime Database, App Check and Web Crypto API into a complete production deployment pipeline	
Dielectric characterization of liquid crystals with terphenyl doping - Research	Dec 2025 - May 2026
- Characterized LC dielectric properties at 3+ terphenyl doping levels using controlled laboratory-based impedance measurements - Mapped permittivity shifts via automated sweeps across 0.2 Hz–8 MHz and a ~20°C temperature window, correlating with doping	
Imager - AI detector for images	Dec 2025 - Jan 2026
- Fused 3 forensic signals (CNN variance, frequency texture, metadata) into a single ensemble inference pipeline via FastAPI backend - Achieved ~85% test accuracy detecting AI-generated images through a multi-signal forensic analysis system built with FastAPI	
FormulaT - Tyre survival prediction tool	Sep 2025 - Oct 2025
- Engineered 20+ race features via FastF1 and trained Random Survival Forests (RSF) to predict per-lap tyre degradation curves - Hit a concordance index of ~0.78 on F1 tyre survival prediction by training on 5 seasons of race telemetry and pit data	
Classif.ai - Classification of satellite imagery	Dec 2024 - Jan 2025
- Utilized Convolutional Neural Networks (CNNs), advanced preprocessing workflows, and Python (TensorFlow, Keras, and NumPy) - Reached ~91% test accuracy classifying 7 landform classes from multispectral satellite imagery using a custom CNN pipeline	

POSITIONS OF RESPONSIBILITY

Outreach and Apogee Coordinator - Team Anant	Mar 2025 - May 2026
- Developed image processing algorithms for 1U/3U CubeSat payloads targeting sub-30m ground resolution for Earth observation - Liaised with 4+ research organizations to drive a planned satellite launch, structuring the data retrieval pipeline end-to-end	
Graphic Design Lead - Students Union Media Team	Mar 2025 - May 2026
- Directed a 10-member design team delivering 30+ assets/semester, creating campaign materials for campus-wide events at BITS Pilani - Grew social media reach by ~25% per semester by running design campaigns across 10+ events including OASIS and APOGEE	

EXTRA CURRICULAR ACTIVITIES

Debating and MUNs
- Achieved top awards and recognition as a leading participant in debate and Model United Nations events at school and university - Chaired 5+ MUNs, evaluating participants and providing structured feedback on argumentation, presentation, and overall performance